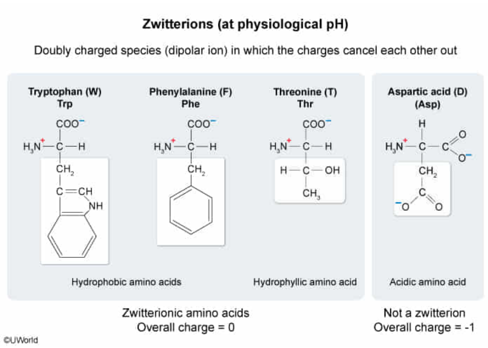
A researcher needs to use an amino acid that is a zwitterion at physiological pH. Which of the following amino acids could NOT be used in the experiment?

- A. D [84%]
- B. W [5%]
- C. F [6%]
- D. T [3%]

Correct
85% answered correctly

Explanation:



Amino acids have three components: an amine group, a carboxyl group, and a side chain. Each component is bonded to the same carbon, known as the α-carbon. Amino acids differ only by the structure of the side chains, which contain different functional groups. The components of the amino acid can take on different charges, depending on the pH of the environment. **Physiological pH** (7.4) is above the **carboxyl group's** pK_a of 2, causing the carboxyl group to become **deprotonated** and gain a **−1 charge**. However, the pH is *below* the **amine** group's pK_a of 9, so the amine is fully **protonated** and has a **+1 charge**. This doubly charged species is known as a **zwitterion**, or a dipolar ion, in which the charges of the amine and carboxyl groups cancel one another out.

Most of the amino acid side chains are uncharged at physiological pH because the side chain functional groups do not have acidic or basic protons. The uncharged amino acid side chains may be hydrophobic or hydrophilic. Tryptophan (Trp, W) and phenylalanine (Phe, F) are uncharged hydrophobic amino acids **(Choices B and C)**. Threonine (Thr, T) is a polar amino acid that is uncharged at physiological pH **(Choice D)**.

Acidic side chains are deprotonated at physiological pH and have a −1 charge, and the basic side chains have a fully protonated nitrogen atom and a +1 charge. Therefore, the overall charge of these amino acids is not zero. **Aspartate (Asp, D)** has an acidic side chain and therefore has a negative charge at physiological pH and **cannot be a zwitterion** at physiological pH.

Educational objective:
At physiological pH (7.4), amino acid backbones have a positively charged amino group (+1 charge) and a negatively charged carboxyl group (−1 charge), and the charges cancel one another out, yielding a net charge of 0. These compounds are known as zwitterions. Acidic and basic amino acid side chains are charged at pH 7.4, so the overall charge of the molecule is not zero and these amino acids are not zwitterions at physiological pH.

Time spent:0 sec
Subject:
Organic Chemistry

QID:402528
System:
Functional Groups and Their Reactions